Communication challenges between individuals with low and high autistic traits: an experimental semiotic approach

Rui Liu (刘睿)[*1,2,] Jan R. Wiersema[2,3], Lara Bardi[1,2]
[*]Corresponding Author: rui.liu@ugent.be
[1] Department of Experimental Psychology, Ghent University, Ghent, Belgium
[2] EXPLORA, Ghent University, Ghent, Belgium
[3] Department of Experimental Clinical and Health Psychology, Ghent University, Ghent, Belgium

1. Introduction

Traditionally, communication difficulties have been conceptualized as arising from inherent impairments in social perception and behavior among individuals with autism. The neurodiversity movement has called for a view flip from deficit-based accounts toward an appreciation of individual differences, emphasizing acceptance of one's self-identity, and empowerment rather than cure or correction. Milton (2012) coined the term Double Empathy Problem (DEP) which suggested that communication breakdowns may often reflect mismatches in lived experience and meaning-making across neurotypes, rather than intrinsic deficits within individuals with autism. Although the DEP has been strongly endorsed by autistic self-reports, empirical evidence supporting the framework remains mixed, due to the complexity of interpersonal exchange, which cannot be fully explained by neurotype match/mismatch alone. Moreover, most studies rely on a signal-based approach—inferring mental states from bodily or linguistic cues—since direct access is not possible. It remains unclear to what extent the "standard meanings" of the signals generalize to individuals on the autism spectrum or reflect researchers' subjective interpretation (Bird & Cook, 2013; Long et al., 2025). From a constructivist perspective, signals do not carry fixed meanings but function as ad-hoc tokens. Meaning is therefore continuously co-constructed between partners as communication unfolds (Brennan et al., 2010; Clark, 1996; Kopp & Krämer, 2021). This process is shaped by global factors preceding communication (e.g., interpersonal asymmetry in autistic traits) and local factors evolving during communication (e.g., ongoing changes of interlocutor's mental states) (Brennan et al., 2010; Davidson, 2005). This perspective invites reconsideration of the double-empathy problem through the lens of inherent interpersonal asymmetry (a global factor) as well as real-time interpersonal adjustment (a local factor). We hypothesize that greater differences in autistic traits may initially hinder communication but can be mitigated through interpersonal adjustments of mental states.

2. Methods

We test this hypothesis within a constructivist framework of communication using an experimental semiotic approach. We formed 27 high-high (HH), 22 low-low (LL), and 27 mixed-trait low-high (LH) AQ dyads (ages 23±4; 106 female, 44 male and 2 other gender) dyads, which is a well acceptable procedure in autism research. The group cutoffs were defined based on the 25th and 75th percentiles (≤15 or ≥25) of the Autism Spectrum Quotient (AQ) scores of 215 adults without diagnosed psychological/neurological disorders. To ensure have sufficient pairs, we additionally assessed AQs of 343 no-diagnoses adults. From the combined sample, 217 participants did not meet the group cutoffs, 122 did not response to us or refuse to participate the behavioral experiment, and 63 participated related experiment and not completed naïve to the topic. Eventually, we managed to recruit the 76 dyads to play the adapted Embodied Communication Game (Fig. 1A&B) (Scott-Phillips et al., 2009), where two players create a novel communication system from scratch. Specifically, in each trial, the players were presented with separate color sets and asked to select the same color without knowing the other's colors. They communicated their intended choices through movement patterns of an icon, received feedback after each trial. They completed 200 rounds across 10 blocks. The task provides an open-ended communicative scenario where solutions, signaling strategies and communicative orders/roles are unconstrained, enabling assessment of asymmetric mental states and spontaneous evolvement of interpersonal adjustment. Mental states of each player were characterized using a Bayesian updating approach that quantified how strong a moving signal was associated with a particular color. Interpersonal adjustment was quantified as the similarity between the adjustor's current mental states and the adjusted's preceding mental states (Fig. 1C).

3. Results

The three types of dyads didn't show any differences in reaction time (the time interval between the presence of the bottles and the movement onset) and moving time (the time interval between the movement onset and the selection of the bottle), which suggests that all participants showed comparable levels of engagement and communicative effort during the task. However, they significantly differed in the levels of interpersonal adjustment ($F = 3.330$, $p = 0.039$, $\omega 2 = 0.030$, Fig. 2A). More precisely, mixed-trait dyads showed a significantly lower level of adjustment than HH dyads ($t = 2.378$, $p = 0.019$, Cohen's $d = 0.462$) and LL dyads ($t = 2.002$, $p = 0.047$, Cohen's $d = 0.407$). No difference was found between HH and LL dyads ($t = -0.271$, $p = 0.787$, Cohen's $d = -0.055$).

Mixed-trait dyads may face greater challenges when implementing adjustment, likely because the players held more divergent mental states in the task than other dyads. This speculation was supported by the observation that mixed-trait dyads elicited more diverse moving patterns in their initial communication, as well as

larger intersubject variances in how they accommodate their mental states towards each other over time. Namely, they showed a greater Simpson's Diversity Index of moving patterns of the first two blocks than HH dyads ($F = 3.255$, $p = 0.041$, $\omega2 = 0.029$, $t = -2.480$, $p = 0.014$, Cohen's $d = -0.477$, Fig. 2B), although there was no difference in the number of moving signals ($F = 1.604$, $p = 0.204$, $\omega2 = 0.008$). Moreover, mixed-trait dyads showed a greater inter-subject variance in the time courses of adjustment, indicated by a larger Dynamic Time Warping (DTW) distance than other dyads ($F = 182.543$, $p < 0.001$, $\omega2 = 0.047$, vs. HH dyads: $t = -17.729$, $p < 0.001$, Cohen's $d = -0.478$, vs. LL dyads: $t = -14.163$, $p < 0.001$, Cohen's $d = -0.420$, Fig. 2C).

Fig. 3A presents a 2D visualization of the adjustment dynamics using Uniform Manifold Approximation and Projection (UMAP). Each data point indicates adjustment dynamics of one participant, colored by the joint correctness in the game. Notably, location of the data points in the UMAP space matches the final joint correctness well. Put differently, how the players adjusted over time predicted how well they performed in the game. As Fig. 3B indicates, those who achieved higher joint correctness started with high levels of adjustment early in the game.

Consistent with some previous reports in this field, we did not find that interpersonal autistic-trait differences directly affect communication outcomes including joint correctness ($z = 1.653$, $p = 0.098$), enjoyment of current interaction ($z = -0.190$, $p = 0.849$), and wiliness for future interaction ($z = 0.925$, $p = 0.925$). Importantly, we hypothesized that greater autistic-trait differences between interlocutors may initially hinder communication but can be mitigated through interpersonal adjustments. In line with this hypothesis, we found that autistic-traits differences indirectly affected communication outcomes through the mediation of interpersonal adjustment. Greater autistic-trait differences between the players within a dyad reduced interpersonal adjustment ($z = -2.825$, $p = 0.005$, Fig. 4A). However, if the players more effectively accommodated their mental states to their co-players', they were more likely to obtain higher joint correctness ($z = 16.023$, $p < 0.001$), experienced greater enjoyment ($z = 4.691$, $p < 0.001$), and became more willing to interact in the future ($z = 2.891$, $p = 0.004$).

In contrast, regardless of their interpersonal autistic-trait differences, longer moving time reduced enjoyment of interaction during the game ($z = -2.376$, $p = 0.017$) and players' willingness for future interaction ($z = -2.604$, $p = 0.009$), although joint correctness remained unaffected ($z = 0.105$, $p = 0.916$). Worth noting is that individual autistic traits neither predicted interpersonal adjustment ($z = 0.720$, $p = 0.472$, Fig 4B), nor affected communication outcomes directly or indirectly. These results highlight the fact that communication is bidirectional.

Conclusions: These findings challenge the notion that communication difficulties are inherent to autism, underscore the bidirectional nature of interactional challenges across neurotypes, and highlight the importance of interpersonal adjustment in bridging divergent minds. This study demonstrates how

experimental semiotics can illuminate the dynamics of intentional communication and can offer novel insights into autism research.

Reference

Bird, G., & Cook, R. (2013). Mixed emotions: The contribution of alexithymia to the emotional symptoms of autism. Translational Psychiatry, 3(7), e285–e285. https://doi.org/10.1038/tp.2013.61

Brennan, S. E., Galati, A., & Kuhlen, A. K. (2010). Chapter 8 - Two Minds, One Dialog: Coordinating Speaking and Understanding. In B. H. Ross (Ed.), Psychology of Learning and Motivation (Vol. 53, pp. 301–344). Academic Press. https://doi.org/10.1016/S0079-7421(10)53008-1

Clark, H. H. (1996). Using Language. Cambridge University Press. https://doi.org/10.1017/CBO9780511620539

Davidson, D. (2005). The Social Aspect of Language. In D. Davidson (Ed.), Truth, Language, and History: Philosophical Essays Volume 5. Oxford University Press. https://doi.org/10.1093/019823757X.003.0008

Kopp, S., & Krämer, N. (2021). Revisiting Human-Agent Communication: The Importance of Joint Co-construction and Understanding Mental States. Frontiers in Psychology, 12, 580955. https://doi.org/10.3389/fpsyg.2021.580955

Long, E. L., Catmur, C., & Bird, G. (2025). The theory of mind hypothesis of autism: A critical evaluation of the status quo. Psychological Review. https://doi.org/10.1037/rev0000532

Milton, D. E. M. (2012). On the ontological status of autism: The 'double empathy problem.' Disability & Society, 27(6), 883–887. https://doi.org/10.1080/09687599.2012.710008

Scott-Phillips, T. C., Kirby, S., & Ritchie, G. R. S. (2009). Signalling signalhood and the emergence of communication. Cognition, 113(2), Article 2. https://doi.org/10.1016/j.cognition.2009.08.009

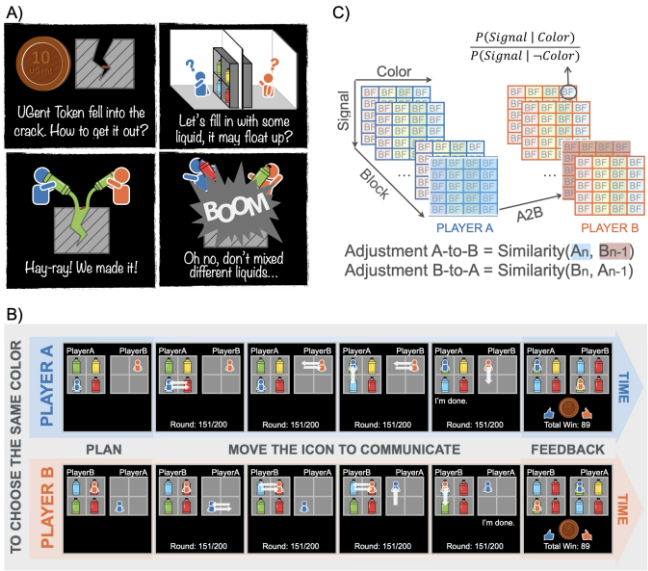

Fig 1. Fig. 1. A–B) Communication scenario and trial structure of the ECG task. A cover story introduced communicative pressure requiring dyadic players to select the same color without knowing their partner's available options. Seated separately, participants controlled an avatar and coordinated exclusively through movement patterns, without visual contact or verbal exchange. No conventional meaning was assigned to movements; signaling systems were constructed de novo. C) Quantification of individual mental states and interpersonal adjustment. Bayes Factors (BFs) indexed the strength of association between specific movement patterns and color choices relative to alternatives. Interpersonal adjustment was operationalized as the similarity between one player's BF matrix in the current block and their partner's BF matrix from the preceding block.

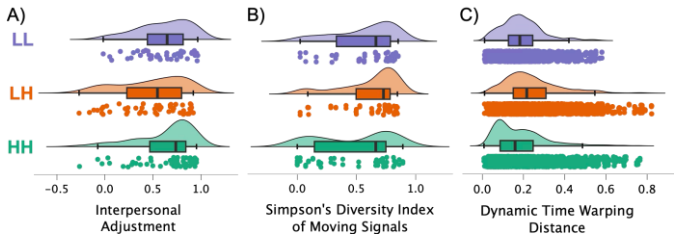

Fig 2. Results of one-way ANOVAs. The low-high (LH, orange) dyads showed A) less interpersonal adjustment; B) greater diversity of moving signals; and C) larger intersubject variance in temporal development of adjustment, as compared to the low-low (LL, purple) and high-high (HH, green) dyads.

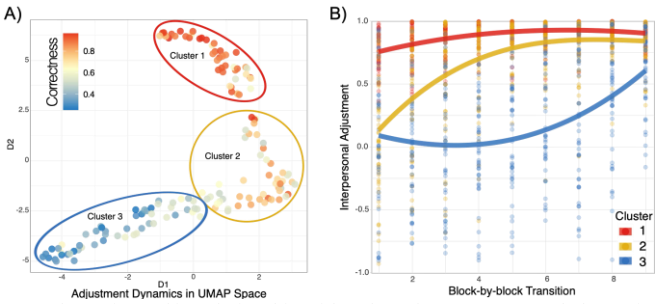

Fig 3. Adjustment dynamics and its relationship with task performance. A) Each data point represented adjustment dynamics per player in 2D UMAP space, colored by correctness they achieved in the game. B) Adjustment dynamics of the players who achieved high (red), medium (yellow) and low (blue) correctness.

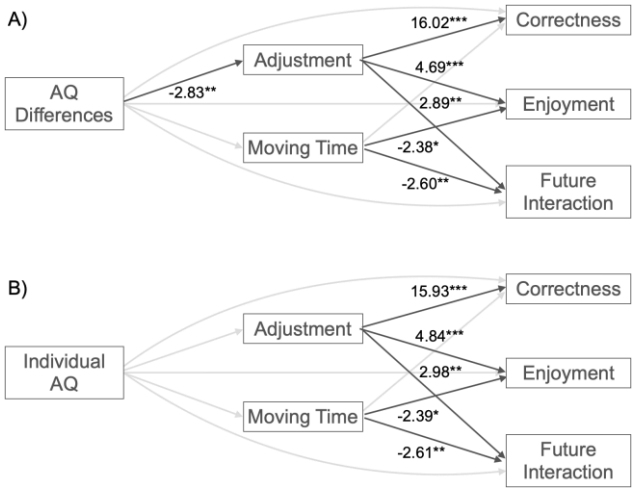

Fig 4. Results of mediation analysis. A) Autistic-trait differences between interlocutors negatively affected communication outcomes (including joint correctness, enjoyment of current interaction, and willingness for future interaction), which was mediated by interpersonal adjustment rather than moving time. B) Communication outcomes were not influenced individual autistic traits. Interpersonal adjustment promotes communication outcomes, but moving time reduced enjoyments of current interaction and willing ness for future interaction. **: $p < 0.01$; ***: $p<0.0$